CHAPTER

02

Kinematics and Dynamics of Particles

Mechanics is the study of motion. **Kinematics** describes *how* things move (position, velocity, acceleration) without asking why. **Dynamics** explains *why* they move, using **forces** and **Newton's laws**. This chapter then adds circular motion and gravitation, the **work–energy** method, and **momentum** and collisions. It carries the most marks in both mid and final exams.

LESSONS IN THIS CHAPTER

2.1 Displacement, Velocity and Acceleration

2.2 Motion with Constant Acceleration and Free Fall

2.3 Projectile Motion

2.4 Forces and Newton's Laws of Motion

2.5 Circular Motion and Gravitation

2.6 Work, Energy and Power

2.7 Linear Momentum, Collisions and Centre of Mass

Freshman A+ · General Physics (Phys 1011)

CHAPTER 2 · LESSON 2.1

Displacement, Velocity and Acceleration

AFTER THIS LESSON YOU CAN

- Distinguish distance from displacement and speed from velocity
- Compute average and instantaneous velocity and acceleration
- Use derivatives for motion given as x(t) or r(t)
- Read motion graphs

2.1.1 Position, distance and displacement

D KEY DEFINITIONS

- **Position** $\vec{r}$: where an object is, measured from a chosen origin.
- **Displacement** $\Delta\vec{r} = \vec{r}_f - \vec{r}_i$: the **change in position**, a **vector** (the straight line from start to finish).
- **Distance**: the **total length of the path** travelled, a **scalar**, never negative.

E DISTANCE ≠ DISPLACEMENT

You walk 20 m east and then come back to where you started. **Distance = 40 m**, but **displacement = 0**. The magnitude of a displacement can never be larger than the distance.

2.1.2 Velocity and speed

$$\vec{v}_{\text{avg}} = \frac{\Delta\vec{r}}{\Delta t} \qquad \text{average speed} = \frac{\text{total distance}}{\text{total time}}$$

Instantaneous velocity is the limit of the average velocity as $\Delta t \to 0$, i.e. the **derivative** of position:

$$\vec{v} = \lim_{\Delta t \to 0} \frac{\Delta\vec{r}}{\Delta t} = \frac{d\vec{r}}{dt}, \qquad v_x = \frac{dx}{dt}$$

Its magnitude is the **instantaneous speed**. On an x–t graph, the velocity is the **slope** of the curve.

2.1.3 Acceleration

$$\vec{a}_{\text{avg}} = \frac{\Delta\vec{v}}{\Delta t}, \qquad \vec{a} = \frac{d\vec{v}}{dt} = \frac{d^2\vec{r}}{dt^2}$$

The SI unit is **m/s²**. **Instantaneous acceleration** is *the limiting value of the average acceleration as the time interval approaches zero*. On a v–t graph, the acceleration is the **slope** and the **area under the curve is the displacement**.

WHEN DOES AN OBJECT ACCELERATE?

Acceleration means the **velocity changes**, and velocity is a vector, so an object accelerates if its **speed changes**, its **direction changes**, or **both**. A car going round a bend at a constant 60 km/h **is accelerating**.

SIGN RULES FOR SPEEDING UP AND SLOWING DOWN

- $\vec{v}$ and $\vec{a}$ in the **same direction** → **speeding up**.
- $\vec{v}$ and $\vec{a}$ in **opposite directions** → **slowing down** (deceleration).
- **Zero acceleration does NOT mean at rest.** It means **constant velocity** (a car cruising at 100 km/h has $a = 0$).
- **Zero velocity does NOT mean zero acceleration.** At the top of a vertical throw, $v = 0$ but $a = g$.
- Average and instantaneous acceleration are **equal only when the acceleration is constant**.

2.1.4 Motion in two dimensions

In 2D (and 3D) we treat each axis separately:

$$\vec{r} = x\hat{i} + y\hat{j}, \qquad \vec{v} = \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j}, \qquad \vec{a} = \frac{dv_x}{dt}\hat{i} + \frac{dv_y}{dt}\hat{j}$$

WORKED EXAMPLE 1: MOTION FROM AN EQUATION (ARBA MINCH EXAM)

A particle moves along x with $x(t) = 10t^2 - 2t^3$ m. Find v and a at $t = 2$ s, and the average velocity between 2 s and 3 s.

$v = \dfrac{dx}{dt} = 20t - 6t^2$ → $v(2) = 40 - 24 = 16$ m/s

$a = \dfrac{dv}{dt} = 20 - 12t$ → $a(2) = 20 - 24 = -4$ m/s²

$x(2) = 40 - 16 = 24$ m, $x(3) = 90 - 54 = 36$ m → $v_{\text{avg}} = \dfrac{36 - 24}{3 - 2} = 12$ m/s

WORKED EXAMPLE 2: 2D MOTION (ASTU EXAM)

$\vec{r}(t) = (2t + 1)\hat{i} + t^2\hat{j}$ m. Find the average velocity from $t = 2$ s to $t = 4$ s and the acceleration at $t = 3$ s.

$\vec{r}(2) = 5\hat{i} + 4\hat{j}$, $\vec{r}(4) = 9\hat{i} + 16\hat{j}$ → $\vec{v}_{\text{avg}} = \dfrac{4\hat{i} + 12\hat{j}}{2} = (2\hat{i} + 6\hat{j})$ m/s

$\vec{v} = 2\hat{i} + 2t\hat{j}$ → $\vec{a} = 2\hat{j}$ m/s² (constant, so it is the same at $t = 3$ s).

WORKED EXAMPLE 3: AVERAGE VELOCITY VS AVERAGE SPEED (ARSI EXAM)

A car travels 60 km east in 0.5 h, then 80 km north in 1.5 h.

Displacement $= \sqrt{60^2 + 80^2} = 100$ km; total time 2.0 h → average **velocity** = 50 km/h.

Distance = 140 km → average **speed** = 70 km/h. The two are **different**.

Exam Practice

11 QUESTIONS · LESSON 2.1

1 Ambo University · Mid Exam

Average velocity and average acceleration of an object do not provide detailed information about the entire motion.

True / False

2 Arba Minch University · Mid Exam

If the acceleration of an object is zero, then that object cannot be moving.

True / False

3 Ambo University · Mid Exam

The direction of the velocity of a body can change while its acceleration is constant.

True / False

4 University of Gondar · Mid Exam

For which of the following is the acceleration always zero?

A) An object moving with constant speed
B) An object moving with constant velocity
C) An object moving along a curve
D) All of the above

5 University of Gondar · Mid Exam

An object moves 20 m east in 30 s and then returns to its starting point in an additional 50 s. Which statement is correct?

A) The average velocity is 0.5 m/s
B) The average speed is 0.5 m/s
C) The average speed is 0
D) The average velocity is 0.25 m/s

6 Arsi University · Mid Exam 2014

A car travels 60 km east in 0.5 h and then 80 km north in 1.5 h. Its average velocity has magnitude:

A) 50 km/h
B) 70 km/h
C) 100 km/h
D) 35 km/h

7 Arsi University · Mid Exam 2014

A particle moves according to $x(t) = (4 + 2t^3)$ m. Its acceleration at $t = 1$ s is:

A) 6 m/s²
B) 2 m/s²
C) 4 m/s²
D) 12 m/s²

8 Arba Minch University · Mid Exam

Which of the following statements is NOT true?

A) When acceleration is constant, instantaneous and average accelerations are equal

B) When velocity and acceleration have opposite signs, the speed increases

C) Free fall is motion under the influence of gravity only

D) Instantaneous velocity is the average velocity over an infinitesimally small time interval

9 Adama Science and Technology University · Mid Exam

A particle moves along x with $x(t) = 3t^2 - 2t + 5$ m. Find (a) the average velocity between $t = 1$ s and $t = 3$ s and (b) the instantaneous velocity at $t = 2$ s.

10 Adama Science and Technology University · Mid Exam

The position of a particle is $\vec{r}(t) = (t^2 + 1)\hat{i} + 3t\hat{j}$ m. Find its velocity vector.

11 Arba Minch University · Mid Exam

Fill in the blank: "_____ is the limiting value of the average acceleration as the change in time approaches zero."

Answers & Explanations

1. True

Averages hide what happens in between: the object may stop, speed up or turn during the interval. Instantaneous values give the details.

2. False

$a = 0$ means **constant velocity**, which may be non-zero (uniform motion in a straight line).

3. True

In projectile motion, $\vec{a} = \vec{g}$ is constant (downward), yet the direction of the velocity changes continuously along the curved path.

4. B

An object moving with constant velocity

Constant velocity means constant speed **and** constant direction, so $a = 0$. Constant speed on a curve still has (centripetal) acceleration.

5. B

The average speed is 0.5 m/s

Distance $= 40$ m in 80 s → average speed $= 0.5$ m/s. Displacement $= 0$ → average velocity $= 0$.

6. A

50 km/h

Displacement $= \sqrt{60^2 + 80^2} = 100$ km over 2.0 h → 50 km/h. (70 km/h would be the average **speed**.)

7. **D**

12 m/s²

$v = 6t^2$, $a = 12t$ → $a(1) = 12$ m/s².

8. **B**

When velocity and acceleration have opposite signs, the speed increases

Opposite signs mean the object is **slowing down**. The speed increases only when v and a have the same sign.

9. **(a) 10 m/s; (b) 10 m/s**

(a) $x(1) = 6$ m, $x(3) = 26$ m → $v_{\text{avg}} = 20/2 = 10$ m/s.
(b) $v = 6t - 2$ → $v(2) = 10$ m/s.

10. **$\vec{v} = (2t\hat{i} + 3\hat{j})$ m/s**

Differentiate each component: $\frac{d}{dt}(t^2 + 1) = 2t$ and $\frac{d}{dt}(3t) = 3$.

11. **Instantaneous acceleration**

$\vec{a} = \lim_{\Delta t \to 0} \Delta\vec{v}/\Delta t = d\vec{v}/dt$.

CHAPTER 2 · LESSON 2.2

Motion with Constant Acceleration and Free Fall

AFTER THIS LESSON YOU CAN

- Use the four kinematic equations
- Choose the right equation from the given and the unknown
- Solve free-fall and vertical-throw problems
- Avoid sign mistakes

2.2.1 The four kinematic equations

When the acceleration is **constant**, motion obeys four equations. (v_0 = initial velocity, v = final velocity, $\Delta x = x - x_0$.)

THE KINEMATIC EQUATIONS (CONSTANT A ONLY)

#	Equation	Missing quantity
1	$v = v_0 + at$	Δx
2	$\Delta x = v_0 t + \frac{1}{2}at^2$	v
3	$v^2 = v_0^2 + 2a\,\Delta x$	t
4	$\Delta x = \frac{v_0 + v}{2}\,t$	a

HOW TO PICK THE EQUATION

List the **five quantities** $v_0,\ v,\ a,\ t,\ \Delta x$. You will know **three**; find the one you need. **Choose the equation that does not contain the quantity you neither know nor need.**

WORKED EXAMPLE 1: BRAKING JET (DILLA EXAM)

A jet lands at 100 m/s and slows down at 4 m/s². How long does it take to stop, and how far does it travel?

Given: $v_0 = 100$ m/s, $v = 0$, $a = -4$ m/s².

Time: $0 = 100 - 4t \Rightarrow t = 25$ s.

Distance: $0 = 100^2 + 2(-4)\Delta x \Rightarrow \Delta x = 10\,000/8 = 1250$ m.

WORKED EXAMPLE 2: FINDING V_0 AND A (DILLA EXAM)

A car decelerates uniformly over 100 m in 10 s and ends with a speed of 2 m/s. Find its initial speed and acceleration.

Equation 4: $100 = \frac{v_0 + 2}{2}(10) \Rightarrow v_0 = 18$ m/s.

Equation 1: $2 = 18 + a(10) \Rightarrow a = -1.6$ m/s².

2.2.2 Free fall

D FREE FALL

Motion **under the influence of gravity alone** (air resistance neglected). Every object, heavy or light, falls with the **same acceleration** $g \approx 9.8$ m/s² (often 10 m/s² in exams), directed **downward**.

Free fall is just constant-acceleration motion with $a = -g$ (taking **up as positive**):

$$v = v_0 - gt, \qquad y = y_0 + v_0 t - \tfrac{1}{2} g t^2, \qquad v^2 = v_0^2 - 2g(y - y_0)$$

F USEFUL RESULTS FOR A VERTICAL THROW FROM THE GROUND (SPEED V_0)

- Time to the top: $t_{\text{up}} = \dfrac{v_0}{g}$; at the top, $v = 0$ but $a = g$.
- Maximum height: $H = \dfrac{v_0^2}{2g}$
- Total time in the air: $T = \dfrac{2v_0}{g}$
- **Symmetry:** it returns to the launch point with the **same speed** v_0, moving downward.
- Dropped from rest from height h: $t = \sqrt{2h/g}$, $v = \sqrt{2gh}$.

✎ WORKED EXAMPLE 3: BALL THROWN UP FROM A BUILDING (ARSI EXAM)

A ball is thrown upward at 10 m/s from the top of a 40 m building ($g = 10$ m/s²). When does it hit the ground?

Origin at the top, up positive: $-40 = 10t - 5t^2 \Rightarrow t^2 - 2t - 8 = 0 \Rightarrow (t-4)(t+2) = 0$.

$t = 4$ **s** (reject the negative root).

! FREE-FALL TRAPS

- **Mass does not matter** (no air resistance): a stone and a piece of paper dropped together land **together**.
- A free-falling body is **not in equilibrium**: it accelerates at g.
- Two balls thrown from a cliff at the same speed, one **up** and one **down**, hit the ground with the **same speed** (energy conservation).
- If one rock is dropped 0.5 s after another, the **distance between them increases** while both fall, because the first rock is always faster.
- Keep one sign convention throughout. With up positive, $a = -g$ for the whole flight, even on the way up.

Exam Practice

13 QUESTIONS · LESSON 2.2

1 Ambo University · Mid Exam

A bike starts from rest and reaches 10 m/s after covering 50 m. Its acceleration is:

A) 0.5 m/s² **B)** 1 m/s²
C) 2 m/s² **D)** 5 m/s²

2 Bahir Dar University · Mid Exam

A car starts from rest and reaches 20.0 m/s in 15.0 s. How far does it travel?

A) 150 m **B)** 300 m
C) 75 m **D)** 200 m

3 Dilla University · Mid Exam 2012

A jet lands at 100 m/s and slows down at a rate of 4 m/s² until it stops. The time needed is:

A) 20 s **B)** 22 s
C) 25 s **D)** 400 s

4 Arsi University · Final Exam 2014

A constant force of 100 N acts for 2 minutes on a 20 kg mass initially at rest. The speed acquired and the distance travelled are:

A) 10 m/s and 600 m **B)** 600 m/s and 36 km
C) 600 m/s and 72 km **D)** 300 m/s and 36 km

5 Ambo University · Mid Exam

A ball is thrown vertically upward at 40 m/s. Its speed after 2 s is ($g = 10$ m/s²):

A) 40 m/s **B)** 20 m/s
C) 60 m/s **D)** 0

6 Bahir Dar University · Mid Exam

An object is thrown vertically upward at 35 m/s. Its velocity after 5 s is ($g = 10$ m/s²):

A) 15 m/s up **B)** 15 m/s down
C) 85 m/s down **D)** 50 m/s up

7 Arsi University · Mid Exam 2014

A ball is thrown upward at 10 m/s from the top of a 40 m building. It reaches the ground after ($g = 10$ m/s²):

A) 2 s **B)** 3 s
C) 4 s **D)** 8 s

8 Dilla University · Mid Exam 2012

A boy drops a stone and a piece of paper from a building at the same time. Neglecting air resistance, which reaches the ground first?

A) The piece of paper
B) Both arrive at the same time
C) The stone
D) It depends on the height

9 Ambo University · Mid Exam

An object that is in free fall is acted upon by the force of gravity alone.

True / False

10 Addis Ababa University · Final 2021

A person at the edge of a cliff throws one ball straight up and an identical ball straight down, both with the same initial speed. Neglecting air resistance, which ball hits the ground with the greater speed?

A) The ball thrown upward
B) The ball thrown downward
C) Neither; they hit at the same speed
D) It is difficult to estimate

11 University of Gondar · Mid Exam

A rock is dropped from a tower, and 0.5 s later a second rock of twice the mass is dropped. Neglecting air resistance:

A) The distance between the rocks increases while both are falling
B) The distance between the rocks stays constant
C) The heavier rock catches up with the first
D) The rocks hit the ground at the same time

12 Dilla University · Mid Exam 2012

A car decelerates uniformly, covering 100 m in 10 s, and ends with a speed of 2 m/s. Find (a) its original speed and (b) its acceleration.

13 Adama Science and Technology University · Exam

A particle is launched vertically upward from the ground with speed v_0. After how long does it return to the ground?

Answers & Explanations

1. B

1 m/s²

$v^2 = v_0^2 + 2a\Delta x \Rightarrow 100 = 2a(50) \Rightarrow a = 1$ m/s².

2. A

150 m

$\Delta x = \frac{v_0 + v}{2} t = \frac{0 + 20}{2}(15) = 150$ m.

3. C

25 s

$0 = 100 - 4t \Rightarrow t = 25$ s.

4. B

600 m/s and 36 km

$a = F/m = 5$ m/s², $t = 120$ s → $v = 5(120) = 600$ m/s; $\Delta x = \frac{1}{2}(5)(120)^2 = 36\,000$ m = 36 km.

5. B

20 m/s

$v = v_0 - gt = 40 - 20 = 20$ m/s (still moving up).

6. B

15 m/s down

$v = 35 - 10(5) = -15$ m/s. The negative sign means it is now moving **down**.

7. C

4 s

$-40 = 10t - 5t^2 \Rightarrow t^2 - 2t - 8 = 0 \Rightarrow t = 4$ s.

8. B

Both arrive at the same time

In free fall every object has the same acceleration g, independent of mass.

9. True

This is the definition of free fall: no air resistance or other forces, only gravity.

10. C

Neither; they hit at the same speed

The upward ball comes back down past the edge with the same speed v_0 it was thrown with, then repeats the downward ball's motion. By energy conservation, $v^2 = v_0^2 + 2gh$ for both.

11. A

The distance between the rocks increases while both are falling

Both accelerate at g, but the first rock always has a 0.5 s head start in speed ($\Delta v = 0.5g$), so the gap keeps growing.

12. (a) 18 m/s; (b) −1.6 m/s²

(a) $100 = \frac{v_0 + 2}{2}(10) \Rightarrow v_0 = 18$ m/s.

(b) $2 = 18 + 10a \Rightarrow a = -1.6$ m/s².

13. $t = 2v_0/g$

$0 = v_0 t - \frac{1}{2}gt^2 = t\left(v_0 - \frac{1}{2}gt\right) \Rightarrow t = 0$ (launch) or $t = 2v_0/g$.

CHAPTER 2 · LESSON 2.3

Projectile Motion

AFTER THIS LESSON YOU CAN

- Treat horizontal and vertical motion independently
- Derive time of flight, maximum height and range
- Solve projectiles launched from a height
- Use the complementary-angle and R–H relations

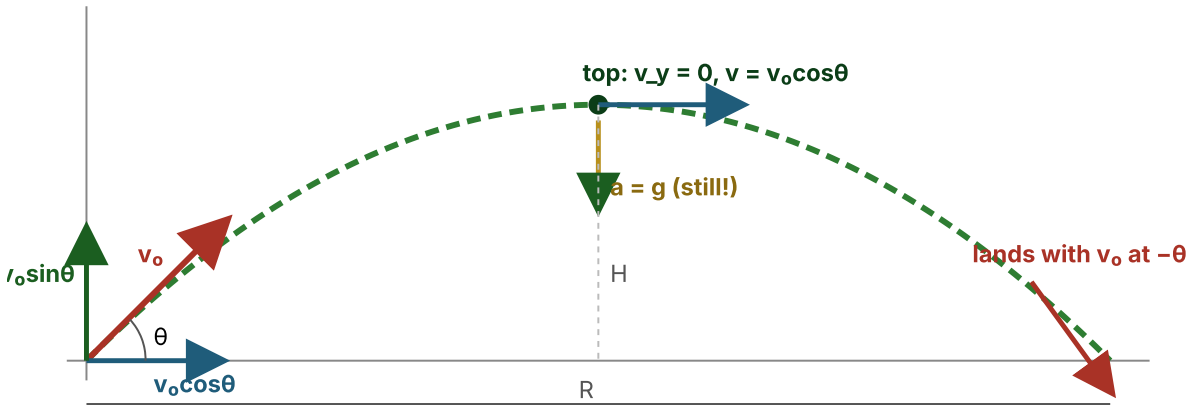
2.3.1 What is a projectile?

A **projectile** is an object thrown into the air that then moves **under gravity alone**. Its path (**trajectory**) is a **parabola**. A **trajectory** is *the path followed by a moving object*.

★ **THE GOLDEN RULE: TWO INDEPENDENT MOTIONS**

	Horizontal (x)	Vertical (y)
Acceleration	$a_x = 0$	$a_y = -g$
Velocity	$v_x = v_0 \cos\theta$ = **constant**	$v_y = v_0 \sin\theta - gt$ (changes)
Position	$x = (v_0 \cos\theta)\, t$	$y = (v_0 \sin\theta)\, t - \frac{1}{2} g t^2$

Horizontal: **constant velocity**. Vertical: **free fall**. The only link between them is **time**.



A projectile launched from level ground

2.3.2 Level-ground formulas

F PROJECTILE LAUNCHED AND LANDING AT THE SAME HEIGHT

$$T = \frac{2v_0 \sin\theta}{g} \qquad H = \frac{v_0^2 \sin^2\theta}{2g} \qquad R = \frac{v_0^2 \sin 2\theta}{g}$$

- **Maximum range** occurs at $\theta = 45^\circ$: $R_{\max} = v_0^2/g$.
- **Complementary angles** (θ and $90^\circ - \theta$, e.g. 30° and 60°) give the **same range** but **different heights** (and different times).
- At the top, $v_y = 0$ and the speed is $v_0 \cos\theta$ (the minimum speed, not zero), while the acceleration is still g.
- **Trajectory equation:** $y = x\tan\theta - \dfrac{g\,x^2}{2v_0^2\cos^2\theta}$, a parabola.

F RELATION BETWEEN RANGE AND HEIGHT

$$\frac{R}{H} = \frac{4}{\tan\theta} \quad\Longrightarrow\quad \tan\theta = \frac{4H}{R}$$

So $R = H \Rightarrow \tan\theta = 4$ ($\theta \approx 76^\circ$); $R = 3H \Rightarrow \tan\theta = \frac{4}{3}$ ($\theta \approx 53^\circ$); $R = 4H \Rightarrow \tan\theta = 1$ ($\theta = 45^\circ$).

WORKED EXAMPLE 1: CANNON ON LEVEL GROUND (AMBO EXAM)

A cannon fires a shell at 20 m/s and 60° ($g = 10$ m/s²). Find the range.

$$R = \frac{v_0^2 \sin 2\theta}{g} = \frac{400 \times \sin 120^\circ}{10} = \frac{400 \times 0.87}{10} = 34.8 \text{ m}.$$

2.3.3 Projectiles launched from a height

The level-ground formulas **do not apply** when the launch and landing heights differ. Go back to the component equations and **solve the vertical equation (a quadratic) for** t.

WORKED EXAMPLE 2: BALL THROWN FROM A CLIFF (AASTU MID EXAM)

A ball is thrown at 20 m/s and 37° above the horizontal from the edge of a 100 m cliff ($\sin 37^\circ = 0.6$, $\cos 37^\circ = 0.8$, $g = 10$ m/s²). Find (a) the time of flight, (b) the horizontal distance, and (c) the impact velocity.

Components: $v_{0x} = 16$ m/s, $v_{0y} = 12$ m/s. Origin at the ground, up positive.

(a) $0 = 100 + 12t - 5t^2 \Rightarrow 5t^2 - 12t - 100 = 0 \Rightarrow t = \dfrac{12 + \sqrt{144 + 2000}}{10} = 5.83$ s.

(b) $x = 16 \times 5.83 = 93.3$ m.

(c) $v_x = 16$ m/s, $v_y = 12 - 10(5.83) = -46.3$ m/s → $v = \sqrt{16^2 + 46.3^2} = 49$ m/s, at $\tan^{-1}(46.3/16) = 70.9^\circ$ **below** the horizontal.

WORKED EXAMPLE 3: FROM A WINDOW (AASTU MID EXAM 2)

A ball is thrown at 25 m/s and 53° above the horizontal from a window and lands 3 s later ($\sin 53^\circ = 0.8$, $g = 10$ m/s²). Find the horizontal distance and the height of the window.

$x = (25 \times 0.6)(3) = 45$ m.

$y = (25 \times 0.8)(3) - 5(3)^2 = 60 - 45 = 15$ m. Since the ball ends at the ground, the window is **15 m** high.

! PROJECTILE TRAPS

- The **horizontal velocity never changes**: it is not zero at the top and does not change by 9.8 m/s each second.
- The **vertical velocity** is zero at the top, but the **acceleration is** g, not zero.
- A bullet fired **horizontally** and a bullet **dropped** from the same height hit the ground **at the same time**. The fired one lands farther away and faster.
- Use $\sin 2\theta$ in the range formula, not $2\sin\theta$.

Exam Practice

15 QUESTIONS · LESSON 2.3

1 Ambo University · Mid Exam

In projectile motion, the horizontal motion has constant acceleration while the vertical motion has constant velocity.

True / False

2 Arba Minch University · Mid Exam

The horizontal velocity of a projectile is 0 m/s at the peak of its trajectory.

True / False

3 Arba Minch University · Mid Exam

The horizontal velocity of a projectile changes by 9.8 m/s each second.

True / False

4 Ambo University · Mid Exam

Which of the following is a non-accelerated motion?

A) The vertical component of projectile motion

B) Circular motion at constant speed

C) The Earth's motion about the Sun

D) The horizontal component of projectile motion

5 Bahir Dar University · Mid Exam

Which of the following is NOT true for a projectile?

A) The horizontal component of its velocity changes with time

B) The only force acting on it is gravity

C) Its acceleration at maximum height is non-zero

D) Its vertical velocity at maximum height is zero

6 Arsi University · Final Exam 2014

Object A is fired horizontally with a very large speed while object B is released from rest at the same height at the same time. Which statement is FALSE?

A) Object B reaches the ground first
B) Object A hits the ground with a larger speed
C) Both reach the ground at the same time
D) Object A covers a larger horizontal distance

7 Ambo University · Mid Exam

A cannon fires a ball at 20 m/s and 60° above the horizontal ($\sin 60^\circ = 0.87$, $g = 10$ m/s²). Its horizontal range on level ground is:

A) 20 m
B) 40 m
C) 17.4 m
D) 34.8 m

8 Arba Minch University · Mid Exam

Two projectiles are fired with the same speed at complementary angles (e.g. 30° and 60°). Which statement is WRONG?

A) Their ranges are equal
B) The maximum range occurs at 45°
C) Their maximum heights are the same
D) The one at 60° stays longer in the air

9 Wachemo University · Mid Exam 2013

A projectile is fired so that its horizontal range equals four times its maximum height. The angle of projection is:

A) 90°
B) 60°
C) 45°
D) 30°

10 Dilla University · Mid Exam 2012

At what angle must a projectile be fired so that its range equals 3 times its maximum height?

11 Adama Science and Technology University · Exam

Find the angle of projection for which the range equals the maximum height.

12 Adama Science and Technology University · Mid Exam

A cannon fires a ball at 10 m/s and 30° above the horizontal on level ground ($g = 10$ m/s²). Find the time of flight, maximum height and range.

13 Adama Science and Technology University · Exam

A projectile is launched at 20 m/s and 53° ($\sin 53^\circ = 0.8$, $g = 10$ m/s²). Find (a) its velocity at maximum height and (b) its position at $t = 0.5$ s.

14 AASTU · Mid Exam 2

A ball is thrown from a window at 25 m/s and 53° above the horizontal and hits the ground 3 s later ($\sin 53^\circ = 0.8$, $\cos 53^\circ = 0.6$, $g = 10$ m/s²). Find (a) the horizontal distance, (b) the height of the window, and (c) the impact speed.

15 University of Gondar · Mid Exam

A ball is thrown at 45° and just clears a wall at horizontal distance A, landing a distance B beyond the wall. Find the height of the wall.

Answers & Explanations

1. **False**

It is the reverse: horizontal → constant **velocity** ($a_x = 0$); vertical → constant **acceleration** ($a_y = -g$).

2. **False**

Only the **vertical** velocity is zero at the peak. The horizontal velocity stays $v_0 \cos\theta$ throughout.

3. **False**

$a_x = 0$, so v_x is constant. It is the **vertical** velocity that changes by 9.8 m/s every second.

4. **D**

The horizontal component of projectile motion

$a_x = 0$ in projectile motion. The other three all have acceleration (gravity, or a change of direction).

5. **A**

The horizontal component of its velocity changes with time

The horizontal velocity is constant; the other statements are all correct.

6. **A**

Object B reaches the ground first

The vertical motions are identical (both start with $v_y = 0$), so they land **together**.

7. **D**

34.8 m

$R = v_0^2 \sin 2\theta / g = 400 \times \sin 120^\circ / 10 = 400 \times 0.87/10 = 34.8$ m.

8. **C**

Their maximum heights are the same

$H \propto \sin^2\theta$, which differs for 30° and 60°. The ranges are equal because $\sin 2\theta$ is the same.

9. **C**

45°

$\tan\theta = 4H/R = 4H/(4H) = 1 \Rightarrow \theta = 45^\circ$.

10. **θ = tan⁻¹(4/3) ≈ 53°**

$\frac{v_0^2 \sin 2\theta}{g} = 3 \cdot \frac{v_0^2 \sin^2\theta}{2g} \Rightarrow 2\sin\theta\cos\theta = \frac{3}{2}\sin^2\theta \Rightarrow \tan\theta = \frac{4}{3}$.

11. **θ = tan⁻¹ 4 ≈ 76°**

$\tan\theta = 4H/R = 4$.

12. **1 s; 1.25 m; 8.66 m**

$v_{0x} = 8.66$ m/s, $v_{0y} = 5$ m/s.

$T = 2v_{0y}/g = 1$ s; $H = v_{0y}^2/2g = 25/20 = 1.25$ m; $R = v_{0x}T = 8.66$ m.

13. **(a) 12î m/s; (b) (6î + 6.75ĵ) m**

$v_{0x} = 12$ m/s, $v_{0y} = 16$ m/s.

(a) At the top $v_y = 0$, so $\vec{v} = 12\hat{i}$ m/s.

(b) $x = 12(0.5) = 6$ m; $y = 16(0.5) - 5(0.25) = 6.75$ m.

14. **(a) 45 m; (b) 15 m; (c) 18.0 m/s**

(a) $x = 15 \times 3 = 45$ m.

(b) $y = 20(3) - 5(9) = 15$ m, so the window is 15 m high.

(c) $v_x = 15$, $v_y = 20 - 30 = -10$ m/s → $v = \sqrt{325} = 18.0$ m/s, at 33.7° below the horizontal.

15. **h = AB/(A + B)**

The range is $R = A + B = v_0^2/g$ (at 45°). The trajectory at 45° is $y = x - \frac{gx^2}{v_0^2} = x - \frac{x^2}{A+B}$.

At $x = A$: $h = A - \frac{A^2}{A+B} = \frac{AB}{A+B}$.

CHAPTER 2 · LESSON 2.4

Forces and Newton's Laws of Motion

AFTER THIS LESSON YOU CAN

- Define force and net force
- Classify forces and name the four fundamental interactions
- State and apply Newton's three laws
- Solve friction, incline, pulley and elevator problems with free-body diagrams

2.4.1 Force and net force

D FORCE

Any **interaction that changes (or tends to change) the motion** of an object: it can start, stop, speed up, slow down or turn a body, and it can also **change its shape or size**. Force is a **vector**; its SI unit is the **newton** (1 N = 1 kg·m/s²).

Net force $\sum \vec{F}$ is the **vector sum of all forces** acting on the object. The object accelerates **only if the net force is not zero**.

2.4.2 Types of force

The four fundamental interactions

All forces in nature come from four basic interactions:

Force	Acts on	Relative strength	Range	Notes
Strong nuclear	quarks; protons and neutrons	strongest (~ 1)	very short (10^{-15} m)	holds the nucleus together
Electromagnetic	electric charges	$\sim 10^{-2}$	infinite	attractive **or** repulsive; behind friction, normal force and tension
Weak nuclear	all particles (in radioactive decay)	$\sim 10^{-6}$	very short (10^{-18} m)	β-decay
Gravitational	all masses	**weakest** ($\sim 10^{-38}$)	**infinite**	**always attractive**

Fundamental interactions are characterized by the **particles that feel the force**, its **relative strength**, its **range**, and the **particles that mediate it**.

M WEAKEST TO STRONGEST: "GOOD WEEKENDS EMBRACE SLEEP"

Gravity < **W**eak < **E**lectromagnetic < **S**trong.

Contact and non-contact forces (module classification)

CONTACT FORCES

Require physical contact: **muscular**, **friction**, **normal**, **applied**, **tension**, **spring** and **air resistance**.

NON-CONTACT (FIELD) FORCES

Act at a distance: **gravitational**, **magnetic** and **electrostatic** forces.

Normal force N: the push of a surface, always **perpendicular to the surface** (for a book pressed against a vertical wall, it points **horizontally out from the wall**). **Tension** T: the pull of a stretched rope or cable along its length.

2.4.3 Newton's three laws

FIRST LAW (LAW OF INERTIA)

Every body continues in its state of rest or of uniform motion in a straight line unless it is compelled to change that state by a net external force. With **no net force**, velocity is **constant** (it may be zero).
Inertia is the tendency of a body to resist changes in its state of motion. **Mass** is the measure of inertia.

INERTIA IN DAILY LIFE

Passengers jerk **backward** when a bus starts, fall **forward** when it brakes suddenly, and slide **outward** (left) when it turns right. A book stays at rest on a table. These are all the first law.

SECOND LAW

The acceleration of a body is directly proportional to the net force on it and inversely proportional to its mass, and it is in the direction of the net force.

$$\sum \vec{F} = m\vec{a} \qquad \left(\sum F_x = ma_x, \quad \sum F_y = ma_y\right)$$

THIRD LAW

For every action there is an equal and opposite reaction: $\vec{F}_{AB} = -\vec{F}_{BA}$.

- Action and reaction **always exist in pairs** and **exist at the same time**; a single isolated force cannot exist.
- They are **equal in magnitude**, **opposite in direction** and act **along the same line**.
- They act on **DIFFERENT bodies**, so they **never cancel** each other.

THE THIRD-LAW CLASSIC

Weight mg (Earth pulls the book) and the normal force (table pushes the book) are **not** an action–reaction pair: they act on the **same** body. The **reaction to the book's weight** is the **book pulling the Earth upward** with force mg.

Weight is the gravitational force: $W = mg$. Mass is the same everywhere; weight depends on g. A man of weight 900 N on Earth ($g = 10$) has mass 90 kg, so on Jupiter ($g \approx 25.9$ m/s²) he weighs about 2330 N.

2.4.4 Friction

D FRICTION

The force between two surfaces in contact that **opposes relative motion** (kinetic) or **impending motion** (static). It acts **parallel to the surface**, and it is proportional to the **normal force**:

$$f_s \leq \mu_s N \qquad f_k = \mu_k N$$

- **Static friction** adjusts itself from zero up to a **maximum** $f_{s,\max} = \mu_s N$. The object starts to slide when the applied force exceeds this.
- **Kinetic friction** acts when the object **slides**; $\mu_k < \mu_s$, so **it is easier to keep something moving than to start it**.
- The coefficients depend on the **nature of the surfaces** and are nearly **independent of the contact area**. Typical values range from 0.03 to 1.0.
- Friction lets us **walk** and lets wheels roll without slipping.

! FRICTION IS NOT ALWAYS MMG, AND NOT ALWAYS EQUAL TO THE WEIGHT

$f = \mu N$, where N is the **normal** force. On a horizontal floor with only horizontal pushes, $N = mg$. But if you push **down** on the object, N (and friction) increase; if you pull **upward** at an angle, N decreases; on an incline, $N = mg\cos\theta$. Static friction is only **as large as needed**, up to $\mu_s N$.

✎ WORKED EXAMPLE 1: FINDING MS AND MK (MODULE)

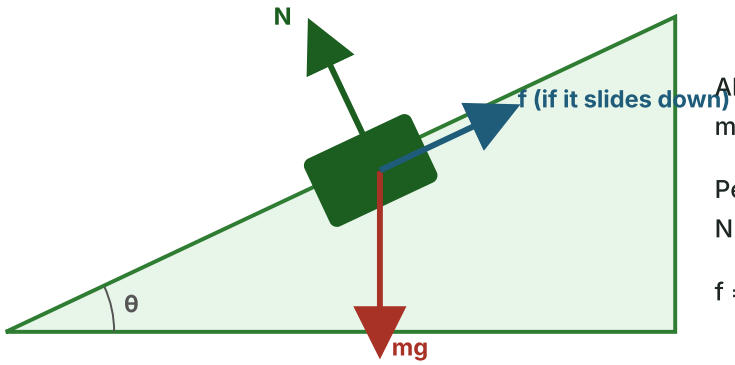
A 25.0 kg block needs 75.0 N to start moving and 60.0 N to keep it moving at constant speed.

$N = mg = 25.0 \times 9.8 = 245$ N → $\mu_s = 75.0/245 = 0.306$; $\mu_k = 60.0/245 = 0.245$.

2.4.5 Solving dynamics problems

★ THE 5-STEP METHOD (MODULE)

1. **Sketch** the situation.
2. Draw a **free-body diagram (FBD)** for **one body at a time**, showing only the forces acting **on** that body.
3. Choose **axes** (on an incline: along and perpendicular to the plane).
4. Write $\sum F_x = ma_x$ and $\sum F_y = ma_y$ for each body.
5. **Solve** for the unknowns.



Along the incline:
mg sinθ − f = ma

Perpendicular:
N = mg cosθ

$f = \mu_k N = \mu_k mg \cos\theta$

Free-body diagram of a block sliding down a rough incline

STANDARD RESULTS (MEMORIZE)

Situation	Result
Block on a **frictionless** incline	$a = g\sin\theta$
Sliding **down** a rough incline	$a = g(\sin\theta - \mu_k\cos\theta)$
Projected **up** a rough incline (slowing)	$a = -g(\sin\theta + \mu_k\cos\theta)$
Slides down at **constant velocity**	$\mu_k = \tan\theta$
Atwood machine ($m_2 > m_1$, frictionless pulley)	$a = \dfrac{(m_2 - m_1)g}{m_1 + m_2}$, $T = \dfrac{2m_1m_2g}{m_1 + m_2}$
Pulling at angle θ, **constant velocity**, friction μ	$F = \dfrac{\mu mg}{\cos\theta + \mu\sin\theta}$
Elevator (apparent weight = scale reading)	up-accelerating: $N = m(g + a)$; down-accelerating: $N = m(g - a)$; free fall: $N = 0$

WORKED EXAMPLE 2: FRICTION ON AN INCLINE (MADDA WALABU EXAM)

A block slides down a 37° incline with acceleration 4 m/s² ($\sin 37^\circ = 0.6$, $\cos 37^\circ = 0.8$, $g = 10$). Find μ_k.

$g\sin\theta - \mu_k g\cos\theta = a \Rightarrow 6 - 8\mu_k = 4 \Rightarrow \mu_k = 0.25.$

WORKED EXAMPLE 3: TWO BLOCKS OVER A PULLEY ON TWO INCLINES (AASTU MID EXAM 2)

$m_1 = 3.5$ kg on a 30° incline and $m_2 = 8$ kg on a 60° incline are joined by a string over a pulley at the top; $\mu_k = 0.25$ for both, $g = 10$. Find the acceleration and tension.

m_2 slides down and m_1 moves up. Driving force: $m_2 g\sin 60^\circ = 69.3$ N.

Opposing: $m_1 g\sin 30^\circ = 17.5$ N; friction $\mu m_1 g\cos 30^\circ = 7.58$ N and $\mu m_2 g\cos 60^\circ = 10.0$ N.

$$a = \frac{69.3 - 17.5 - 7.58 - 10.0}{3.5 + 8} = \frac{34.2}{11.5} = 2.97 \text{ m/s}^2$$

For m_1: $T = 17.5 + 7.58 + 3.5(2.97) = 35.5$ N.

DYNAMICS TRAPS

- In a truck that accelerates **east** with a crate that does not slide, static friction on the crate points **east**: it is the force that accelerates the crate.
- A spring balance in an elevator accelerating **downward** reads **less** than the true weight; accelerating upward, **more**.
- An object in **equilibrium** has $\sum \vec{F} = 0$: it is either at rest or moving with constant velocity.
- **Kinematics** studies motion without its causes; **dynamics** studies motion together with its causes (forces).

Exam Practice

24 QUESTIONS · LESSON 2.4

1 Arba Minch University · Mid Exam

Newton's second law of motion is sometimes called the law of inertia.

True / False

2 Arba Minch University · Mid Exam

Dynamics is the study of the motion of objects without reference to the cause of the motion.

True / False

3 Arba Minch University · Mid Exam

If a system is in equilibrium, then the sum of all forces acting on it must be zero.

True / False

4 Arba Minch University · Mid Exam

Kinetic friction arises when the object is in motion on the surface.

True / False

5 Madda Walabu University · Final Exam

The force of kinetic friction is less than the maximum value of static friction between an object and a surface.

True / False

6 Arba Minch University · Mid Exam

If an object is pushed against a surface, the frictional force increases and can become more than the weight of the object.

True / False

7 AASTU · Mid Exam 2

The "reaction" force does not cancel the "action" force because:

A) The action force is greater than the reaction force
B) They act on different bodies
C) They are in the same direction
D) The reaction exists only after the action is removed

8 Addis Ababa University · Final 2021

Which of the following is true about action and reaction forces?

A) They always act on a single body
B) A body experiences no net force because of them
C) They exist together
D) They have different lines of action

9 Arsi University · Final Exam 2014

Which of the following is NOT true according to Newton's laws of motion?

A) To every action there is an equal and opposite reaction
B) A body is in equilibrium if it is at rest or in uniform motion
C) A freely falling object is in gravitational equilibrium
D) Force equals mass times acceleration

10 Haramaya University · Mid Exam

Arrange the fundamental forces from the weakest to the strongest:

A) Gravitational → electromagnetic → weak → strong
B) Weak → gravitational → electromagnetic → strong
C) Gravitational → weak nuclear → electromagnetic → strong nuclear
D) Electromagnetic → gravitational → weak → strong

11 Arba Minch University · Mid Exam

Which fundamental force is long-range, always attractive and the weakest?

A) Electromagnetic force
B) Strong nuclear force
C) Weak nuclear force
D) Gravitational force

12 Haramaya University · Mid Exam

Which of the following is a non-contact force?

A) Gravitational force
B) Muscular force
C) Air resistance
D) Applied force

13 Bahir Dar University · Mid Exam

A book is pressed flat against a vertical wall. The normal force of the wall on the book points:

A) downward
B) upward
C) out from the wall
D) into the wall

14 Ambo University · Mid Exam

A man weighs 900 N on Earth ($g = 10$ m/s²). What is his weight on a planet where $g = 30$ m/s²?

A) 90 N
B) 300 N
C) 900 N
D) 2700 N

15 Arsi University · Mid Exam 2014

A 2 kg object has acceleration $\vec{a} = (30\hat{i} + 40\hat{j})$ m/s². The magnitude of the net force is:

A) 70 N
B) 140 N
C) 100 N
D) 50 N

16 Addis Ababa University · Final 2021

A crate sits in the centre of a flatbed truck. As the truck accelerates east, the crate moves with it without sliding. The friction force on the crate is directed:

A) West
B) East
C) There is no friction because the crate isn't sliding
D) It is difficult to estimate

17 Addis Ababa University · Final 2021

A body hangs from a spring balance attached to the ceiling of an elevator that is accelerating downward. The balance reads:

A) More than the weight on the ground
B) Equal to the weight on the ground
C) The net force on the body
D) Less than the weight on the ground

18 Arba Minch University · Mid Exam

A 40 kg table rests on a floor with $\mu_s = 0.5$ ($g = 9.8$ m/s²). The greatest horizontal force that can be applied without moving it is:

A) 392 N
B) 196 N
C) 98 N
D) 20 N

19 Madda Walabu University · Final Exam

A block slides down a 37° incline with acceleration 4 m/s² ($\sin 37^\circ = 0.6$, $\cos 37^\circ = 0.8$, $g = 10$). The coefficient of kinetic friction is:

A) 0.25
B) 0.5
C) 0.75
D) 0.2

20 Bahir Dar University · Mid Exam

A block of mass m is pulled at constant speed by a force F at angle θ above the horizontal, with friction coefficient μ. Then F equals:

A) μmg **B)** $\mu mg/\cos\theta$

C) $\mu mg\cos\theta$ **D)** $\mu mg/(\cos\theta + \mu\sin\theta)$

21 Arba Minch University · Mid Exam

A block slides down an incline of angle θ with acceleration $g/3$. Find the coefficient of kinetic friction.

22 Addis Ababa University · Final 2021

A cube slides down a rough incline (angle θ) at constant velocity. It is then projected **up** the same incline with speed V_0. How far up does it go before stopping?

23 AASTU · Mid Exam 2

Blocks $m_1 = 3.5$ kg (on a 30° incline) and $m_2 = 8$ kg (on a 60° incline) are connected over a frictionless pulley; $\mu_k = 0.25$, $g = 10$ m/s². Find (a) the acceleration, (b) the tension, and (c) the direction of motion.

24 Adama Science and Technology University · Mid Exam

Two masses m_1 and m_2 ($m_2 > m_1$) hang over a frictionless pulley (an Atwood machine). Find the acceleration and the tension.

Answers & Explanations

1. **False**

The **first** law is the law of inertia. The second law is $\sum \vec{F} = m\vec{a}$.

2. **False**

That is **kinematics**. Dynamics studies motion together with its causes (forces).

3. **True**

Equilibrium means $\vec{a} = 0$, so $\sum \vec{F} = m\vec{a} = 0$.

4. **True**

Kinetic (sliding) friction acts on surfaces sliding over each other; static friction acts when there is no relative motion.

5. **True**

$\mu_k < \mu_s$, so $f_k = \mu_k N < \mu_s N = f_{s,\max}$. Starting motion takes more force than maintaining it.

6. True

Pushing increases the **normal force**, and $f = \mu N$. For example, a block pressed against a wall can be held by a friction force that balances its weight; with a larger push, the maximum available friction exceeds mg.

7. B

They act on different bodies

Forces can only cancel if they act on the **same** body. An action–reaction pair always acts on two different bodies.

8. C

They exist together

They come in simultaneous pairs, with equal magnitude, opposite direction and the same line of action, on different bodies.

9. C

A freely falling object is in gravitational equilibrium

A freely falling object accelerates at g, so it is **not** in equilibrium.

10. C

Gravitational → weak nuclear → electromagnetic → strong nuclear

Gravity is by far the weakest; the strong nuclear force is the strongest.

11. D

Gravitational force

Gravity has infinite range, is always attractive and is the weakest interaction. The electromagnetic force can also repel.

12. A

Gravitational force

Gravity acts at a distance. The others need contact.

13. C

out from the wall

The normal force is perpendicular to the surface and points away from it (the wall pushes the book outward).

14. D

2700 N

$m = 900/10 = 90$ kg; $W = 90 \times 30 = 2700$ N. His mass stays 90 kg.

15. C

100 N

$\vec{F} = m\vec{a} = (60\hat{i} + 80\hat{j})$ N, so $|\vec{F}| = \sqrt{3600 + 6400} = 100$ N.

16. B

East

The crate accelerates east, so the net force on it must point east. The only horizontal force is **static friction**, so it points east.

17. D

Less than the weight on the ground

$mg - T = ma \Rightarrow T = m(g - a) < mg.$

18. B

196 N

$f_{s,\max} = \mu_s mg = 0.5 \times 40 \times 9.8 = 196$ N.

19. A

0.25

$g\sin\theta - \mu g\cos\theta = a \Rightarrow 6 - 8\mu = 4 \Rightarrow \mu = 0.25.$

20. D

$\mu mg/(\cos\theta + \mu\sin\theta)$

Horizontal: $F\cos\theta = \mu N$. Vertical: $N = mg - F\sin\theta$. So $F\cos\theta = \mu(mg - F\sin\theta) \Rightarrow F = \dfrac{\mu mg}{\cos\theta + \mu\sin\theta}.$

21. $\mu_k = \tan\theta - 1/(3\cos\theta)$

$mg\sin\theta - \mu_k mg\cos\theta = m\frac{g}{3} \Rightarrow \mu_k = \dfrac{\sin\theta - 1/3}{\cos\theta} = \tan\theta - \dfrac{1}{3\cos\theta}.$

22. $d = V_0^2/(4g\sin\theta)$

Constant velocity down gives $\mu_k = \tan\theta$.

Going up, both gravity and friction act down the slope: $a = -g(\sin\theta + \tan\theta\cos\theta) = -2g\sin\theta.$

Then $0 = V_0^2 - 2(2g\sin\theta)d \Rightarrow d = \dfrac{V_0^2}{4g\sin\theta}.$

23. (a) 2.97 m/s²; (b) 35.5 N; (c) m_2 down its incline

(a) $a = \dfrac{m_2 g\sin 60^\circ - m_1 g\sin 30^\circ - \mu m_1 g\cos 30^\circ - \mu m_2 g\cos 60^\circ}{m_1 + m_2} = \dfrac{69.3 - 17.5 - 7.58 - 10.0}{11.5} = 2.97$ m/s².

(b) $T = m_1 g\sin 30^\circ + \mu m_1 g\cos 30^\circ + m_1 a = 17.5 + 7.58 + 10.4 = 35.5$ N.

(c) m_2 moves down the 60° incline and m_1 moves up the 30° incline.

24. $a = (m_2 - m_1)g/(m_1 + m_2)$; $T = 2m_1m_2g/(m_1 + m_2)$

m_2: $m_2 g - T = m_2 a$. m_1: $T - m_1 g = m_1 a$. Adding gives a; substituting back gives $T = m_1(g + a) = \dfrac{2m_1 m_2 g}{m_1 + m_2}.$

CHAPTER 2 · LESSON 2.5

Circular Motion and Gravitation

AFTER THIS LESSON YOU CAN

- Describe uniform circular motion with period, frequency and angular speed
- Compute centripetal acceleration and force
- Apply Newton's law of universal gravitation
- State Kepler's three laws and analyse satellite orbits and weightlessness

2.5.1 Uniform circular motion

Uniform circular motion (UCM) is motion in a circle at **constant speed**. The speed is constant but the **direction of the velocity changes all the time**, so the object **is accelerating**.

F DESCRIBING CIRCULAR MOTION

Quantity	Formula	Unit
Period (time for one revolution)	$T = \dfrac{2\pi r}{v}$	s
Frequency (revolutions per second)	$f = \dfrac{1}{T}$	Hz = s^{-1}
Angular speed	$\omega = \dfrac{2\pi}{T} = 2\pi f$	rad/s
Linear (tangential) speed	$v = \omega r = 2\pi r f$	m/s
Centripetal acceleration	$a_c = \dfrac{v^2}{r} = \omega^2 r$	m/s^2
Centripetal force	$F_c = \dfrac{mv^2}{r} = m\omega^2 r$	N

★ DIRECTION IS EVERYTHING

- The velocity is **tangent** to the circle.
- The centripetal acceleration (and the net force) points **toward the centre**, perpendicular to the velocity, so it changes the **direction**, not the speed.
- The **magnitude** of a_c is constant, but its **direction keeps changing**, so $\vec{a}_c$ is **not a constant vector**.
- If the speed also changes (non-uniform circular motion), there is an extra **tangential acceleration** $a_t = dv/dt$. Total: $a = \sqrt{a_c^2 + a_t^2}$.

D CENTRIPETAL FORCE IS NOT A NEW KIND OF FORCE

"Centripetal" means **centre-seeking**. It is the **name of the job**, done by a real force: **tension** (stone on a string), **friction** (car on a flat curve), **gravity** (Moon round the Earth), the **normal force** (banked road, loop) or the **electric force** (electron round a nucleus). Never draw an extra "centripetal force" on a free-body diagram.

F **STANDARD CIRCULAR-MOTION RESULTS**

Situation	Result
Car on a **flat** curve (friction supplies F_c)	$v_{\max} = \sqrt{\mu_s g r}$
Banked curve, no friction needed	$\tan\theta = \dfrac{v^2}{rg}$
Vertical circle (string), minimum speed at the top	$v_{\text{top}} = \sqrt{gR}$ (tension zero there)
Tension at the **bottom** of a vertical circle	$T = mg + \dfrac{mv^2}{R}$
Conical pendulum / stone whirled at angle θ	$\tan\theta = \dfrac{v^2}{rg}$

WORKED EXAMPLE 1: BALL ON A STRING (HAWASSA EXAM)

A ball on a 1.0 m string turns at 360 rev/min in a horizontal circle. Find its speed and centripetal acceleration.

$f = 360/60 = 6$ rev/s → $v = 2\pi r f = 2\pi(1)(6) = 12\pi = 37.7$ m/s.

$a_c = v^2/r = (12\pi)^2/1 = 144\pi^2 = 1.42 \times 10^3$ m/s² (about 145 g).

WORKED EXAMPLE 2: CAR ON A FLAT CURVE

A 1200 kg car rounds a flat curve of radius 50 m; $\mu_s = 0.8$, $g = 10$ m/s². Find the maximum safe speed.

Friction supplies the centripetal force: $\mu_s mg = \dfrac{mv^2}{r} \Rightarrow v = \sqrt{\mu_s g r} = \sqrt{0.8 \times 10 \times 50} = 20$ m/s (72 km/h). **The mass cancels.**

2.5.2 Newton's law of universal gravitation

D **LAW OF UNIVERSAL GRAVITATION**

Every particle in the universe attracts every other particle with a force that is directly proportional to the product of their masses and inversely proportional to the square of the distance between their centres.

$$F = G\frac{m_1 m_2}{r^2}, \qquad G = 6.67 \times 10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2$$

- Gravity is **always attractive**, acts **along the line joining the centres**, and has **infinite range**.
- The force on each body is the **same size** (Newton's third law): the Earth pulls you exactly as hard as you pull the Earth.
- It is a **universal property of all matter** (anything with mass), not only of large bodies.
- **Inverse-square law:** doubling the distance makes the force **4 times smaller**; tripling it makes it 9 times smaller.

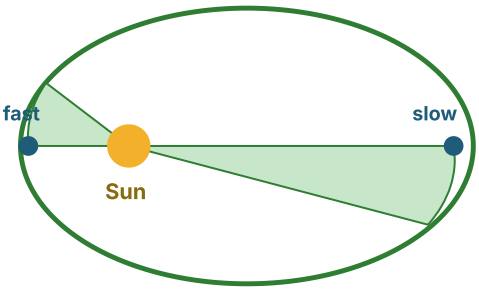
F **GRAVITY NEAR AND ABOVE A PLANET**

- At the surface: $g = \dfrac{GM}{R^2}$ (Earth: $g \approx 9.8$ m/s², $M_E = 5.97 \times 10^{24}$ kg, $R_E = 6.37 \times 10^{6}$ m).
- At height h: $g_h = \dfrac{GM}{(R+h)^2} = g\left(\dfrac{R}{R+h}\right)^2$. At $h = R$: $g_h = g/4$.
- **Escape speed:** $v_{\text{esc}} = \sqrt{\dfrac{2GM}{R}} \approx 11.2$ km/s for Earth.

2.5.3 Kepler's laws of planetary motion

D **THE THREE LAWS**

1. **Law of orbits:** every planet moves in an **ellipse** with the Sun at **one focus**.
2. **Law of areas:** the line from the Sun to a planet **sweeps out equal areas in equal times**. So a planet moves **fastest when closest** to the Sun (perihelion) and slowest when farthest (aphelion).
3. **Law of periods:** $T^2 \propto r^3$, i.e. $\dfrac{T^2}{r^3}$ is **the same for all planets** of the Sun.



Kepler's second law

2.5.4 Satellites and weightlessness

For a satellite in a circular orbit of radius r, **gravity supplies the centripetal force**:

$$\frac{GMm}{r^2} = \frac{mv^2}{r} \quad \Longrightarrow \quad v = \sqrt{\frac{GM}{r}}, \qquad T = 2\pi\sqrt{\frac{r^3}{GM}}$$

★ **WHAT THESE FORMULAS TELL YOU**

- The orbital speed and period **do not depend on the satellite's mass**, only on the planet's mass M and the radius r.
- A **higher** orbit is **slower** and has a **longer period** (this is Kepler's third law: $T^2 = \dfrac{4\pi^2}{GM} r^3$).
- A **geostationary** satellite has $T = 24$ h and orbits above the equator at about 36 000 km altitude, so it stays above the same point on Earth (used for TV and communications).

D **WEIGHTLESSNESS**

Astronauts in orbit are **not** beyond gravity: at the ISS altitude g is still about 90 % of its surface value. They feel **weightless** because they and their spacecraft are **falling together** (free fall) around the Earth, so nothing pushes up on them: the **apparent weight** (normal force) is **zero**. The same happens in a freely falling elevator.

CIRCULAR MOTION AND GRAVITATION TRAPS

- An object in UCM is **accelerating** even though its speed is constant.
- If the string breaks, the stone flies off **along the tangent**, not outward along the radius.
- The "outward pull" felt in a turning car is **inertia** (first law), not a real outward force.
- In Kepler's second law the planet covers a **longer** arc when **closer** to the Sun in the same time, because it is moving **faster**.
- The **period** of a satellite depends on the orbit radius and the planet's mass, **not** on the satellite's own mass.

Exam Practice

12 QUESTIONS · LESSON 2.5

1 Hawassa University · Mid Exam

Which of the following statements about uniform circular motion is NOT correct?

A) The centripetal acceleration is constant throughout the motion
B) Tangential acceleration is responsible for a change of speed
C) Radial acceleration is responsible for a change of direction
D) The velocity is always tangent to the path

2 Ambo University · Mid Exam

An object moving in a circle at constant speed has zero acceleration.

True / False

3 Arba Minch University · Mid Exam

A stone tied to a string is whirled in a horizontal circle. If the string suddenly breaks, the stone moves:

A) Toward the centre
B) Radially outward
C) Along the tangent to the circle at that point
D) In a spiral

4 Hawassa University · Mid Exam

A ball attached to a 1.0 m string rotates in a horizontal circle at 360 rev/min. Find (a) its speed and (b) its centripetal acceleration.

5 Bahir Dar University · Mid Exam

Which of the following is NOT a correct statement about the gravitational force between two bodies?

A) It is always attractive
B) It is exerted only by small-sized particles
C) Its magnitude is the same on each body
D) It acts along the line joining their centres

6 Arba Minch University · Mid Exam

According to Kepler's third law, which ratio is the same for all planets orbiting the Sun?

A) T/R
B) T^2/R
C) T^2/R^3
D) T^3/R^2

7 Wolaita Sodo University · Mid Exam

Which statement about gravitation is correct?

A) The gravitational force is a universal property of matter due to its mass
B) In equal time intervals a planet covers a larger distance when it is farther from the Sun
C) The period of a satellite depends on its own mass
D) Gravitational force can be attractive or repulsive

8 Serway / OpenStax-style problem

The distance between two bodies is tripled. The gravitational force between them becomes:

A) 3 times larger
B) 1/3 as large
C) 9 times larger
D) 1/9 as large

9 Serway / OpenStax-style problem

At what height above the Earth's surface is g one quarter of its surface value? (R = Earth's radius)

A) $R/4$
B) $R/2$
C) $2R$
D) R

10 Model question

Astronauts on the International Space Station float because:

A) There is no gravity in space
B) They are too far from the Earth to feel gravity
C) The Moon's gravity cancels the Earth's gravity
D) They and the station are in continuous free fall around the Earth

11 Serway / OpenStax-style problem

A satellite moves in a circular orbit at a height equal to the Earth's radius ($R = 6.4 \times 10^6$ m, $g = 9.8$ m/s²). Find its orbital speed and period.

12 Model question

A 1000 kg car takes a flat curve of radius 40 m. The coefficient of static friction is 0.4 ($g = 10$ m/s²). What is the maximum speed without skidding? Does it depend on the mass?

Answers & Explanations

1. **A**

The centripetal acceleration is constant throughout the motion

Only the **magnitude** of the centripetal acceleration is constant. Its direction always points to the centre, so it keeps changing as the object moves round.

2. **False**

The direction of the velocity changes, so there is a centripetal acceleration v^2/r toward the centre.

3. **C**

Along the tangent to the circle at that point

With no net force, the stone continues with the velocity it had at that instant (first law), which is tangent to the circle.

4. **(a) 12π ≈ 37.7 m/s; (b) 144π² ≈ 1.42 × 10³ m/s²**

$f = 6$ rev/s, $v = 2\pi r f = 12\pi$ m/s; $a_c = v^2/r = 144\pi^2$ m/s².

5. **B**

It is exerted only by small-sized particles

Gravity acts between **all** bodies with mass, large or small. The other three statements are true.

6. **C**

T^2/R^3

$T^2 \propto R^3$, so T^2/R^3 is constant for every planet of the same star.

7. **A**

The gravitational force is a universal property of matter due to its mass

B is backwards (the planet is fastest when closest); C is false (T depends only on r and M); D is false (gravity only attracts).

8. **D**

1/9 as large

$F \propto 1/r^2$, so tripling r divides F by $3^2 = 9$.

9. **D**

R

$$g_h = g\left(\frac{R}{R+h}\right)^2 = \frac{g}{4} \Rightarrow R + h = 2R \Rightarrow h = R.$$

10. **D**

They and the station are in continuous free fall around the Earth

Gravity at that altitude is still about 90 % of g. It provides the centripetal force for the orbit, and since everything falls together, the apparent weight is zero.

11. **v ≈ 5.6 km/s; T ≈ 4.0 h**

$r = 2R$ and $GM = gR^2$. So $v = \sqrt{GM/r} = \sqrt{gR/2} = \sqrt{9.8 \times 6.4 \times 10^6/2} = 5.6 \times 10^3$ m/s.
$T = 2\pi r/v = 2\pi(1.28 \times 10^7)/5600 = 1.44 \times 10^4$ s ≈ 4.0 h.

12. **12.6 m/s; no**

$\mu_s mg = mv^2/r \Rightarrow v = \sqrt{\mu_s g r} = \sqrt{0.4 \times 10 \times 40} = \sqrt{160} = 12.6$ m/s. The mass cancels.

CHAPTER 2 · LESSON 2.6

Work, Energy and Power

AFTER THIS LESSON YOU CAN

- Calculate the work done by a constant and a variable force
- Apply the work–energy theorem
- Distinguish conservative and non-conservative forces
- Use conservation of mechanical energy and compute power

2.6.1 Work done by a constant force

D WORK

Work is done when a **force causes a displacement**. For a constant force:

$$W = \vec{F} \cdot \vec{d} = Fd\cos\theta = F_x\Delta x + F_y\Delta y$$

where θ is the angle between $\vec{F}$ and $\vec{d}$. Work is a **scalar**; its unit is the **joule** (1 J = 1 N·m), with dimensions $[ML^2T^{-2}]$.

★ THE SIGN OF WORK

Angle θ	Work	Example
0° ≤ θ < 90°	**positive** (energy given to the body)	pushing a cart forward
θ = 90°	**zero**	the normal force on a sliding box; gravity on horizontal motion; the centripetal force in UCM
90° < θ ≤ 180°	**negative** (energy taken away)	kinetic friction; gravity on a rising ball

Three conditions are needed for work: a force, a displacement, and a component of the force **along** the displacement. Holding a heavy bag still, or carrying it horizontally at constant velocity, does **no work** on the bag.

✎ WORKED EXAMPLE 1: WORK FROM COMPONENTS (HARAMAYA EXAM)

A constant force $\vec{F} = (3\hat{i} + 4\hat{j})$ N moves a particle from P(2, 6) m to Q(14, 1) m. Find the work.

$\Delta\vec{r} = (12\hat{i} - 5\hat{j})$ m → $W = (3)(12) + (4)(-5) = 36 - 20 = 16$ J.

2.6.2 Work done by a variable force

If the force changes with position, the work is the **area under the F–x graph**:

$$W = \int_{x_i}^{x_f} F(x)\,dx$$

Spring (Hooke's law): $F_s = -kx$. The work needed to stretch or compress a spring by x from its natural length is

$$W = \tfrac{1}{2}kx^2$$

✎ **WORKED EXAMPLE 2: STRETCHING A SPRING (AMBO EXAM)**

How much work stretches a spring ($k = 125$ N/m) by 2.0 cm? $W = \tfrac{1}{2}(125)(0.020)^2 = 0.025$ J.

2.6.3 Kinetic energy and the work–energy theorem

D **WORK–ENERGY THEOREM**

The net work done on an object equals the change in its kinetic energy.

$$W_{\text{net}} = \Delta K = \tfrac{1}{2}mv_f^2 - \tfrac{1}{2}mv_i^2, \qquad K = \tfrac{1}{2}mv^2 = \frac{p^2}{2m}$$

- Positive net work → the object **speeds up**; negative → it **slows down**; zero → its **speed is unchanged** (the direction may still change, as in UCM).
- $K \propto v^2$: **double the speed → 4 × the kinetic energy**; halve the speed → ¼ of it. Double the **momentum** → also 4 × the kinetic energy.

✎ **WORKED EXAMPLE 3: APPLIED FORCE AND FRICTION (ARBA MINCH EXAM)**

A 5 kg box starts from rest and is pushed 10 m across a floor by a horizontal 20 N force; $\mu_k = 0.2$, $g = 10$ m/s². Find the work done by each force and the final speed.

$W_F = 20 \times 10 = 200$ J. Friction: $f = \mu mg = 10$ N → $W_f = -10 \times 10 = -100$ J. Gravity and normal force: $W = 0$ (perpendicular).

$W_{\text{net}} = 100$ J $= \tfrac{1}{2}(5)v^2 \Rightarrow v = \sqrt{40} = 6.3$ m/s.

2.6.4 Potential energy and conservative forces

D CONSERVATIVE FORCES

Gravity, the spring force, the electric force.

- The work done **depends only on the start and end points**, not on the path.
- The work around **any closed path is zero**.
- A **potential energy** can be defined: $W_c = -\Delta U$.

! NON-CONSERVATIVE FORCES

Friction, air resistance, a push by your hand.

- The work **depends on the path** (a longer path means more friction work).
- They **convert** mechanical energy into heat, sound, etc.
- No potential energy can be defined for them.

F POTENTIAL ENERGIES

- **Gravitational (near the surface):** $U_g = mgh$ (with h measured from any chosen reference level).
- **Elastic (spring):** $U_s = \frac{1}{2}kx^2$
- **Gravitational (general):** $U = -\dfrac{GMm}{r}$

2.6.5 Conservation of mechanical energy

★ THE ENERGY METHOD

Mechanical energy $E = K + U$. If **only conservative forces do work**:

$$K_i + U_i = K_f + U_f$$

With friction (or another non-conservative force):

$$K_i + U_i + W_{nc} = K_f + U_f \qquad (W_{\text{friction}} = -f_k d)$$

The energy method ignores the path and the direction, so it is often **much faster** than using forces and kinematics.

F RESULTS YOU GET FROM ENERGY CONSERVATION

- Fall from rest through height h: $v = \sqrt{2gh}$ (independent of mass and of the path, e.g. a frictionless slide).
- Pendulum released from height h_0: speed at height h is $v = \sqrt{2g(h_0 - h)}$.
- Spring launches a mass: $\frac{1}{2}kx^2 = \frac{1}{2}mv^2 \Rightarrow v = x\sqrt{k/m}$.

✎ WORKED EXAMPLE 4: PENDULUM (ARBA MINCH EXAM)

A pendulum bob of length L is released from a height $L/4$ above its lowest point. Find its speed when it is at height $L/8$.

$Mg\frac{L}{4} = \frac{1}{2}Mv^2 + Mg\frac{L}{8} \Rightarrow \frac{1}{2}v^2 = g\frac{L}{8} \Rightarrow v = \frac{1}{2}\sqrt{gL}$.

WORKED EXAMPLE 5: SPRING LAUNCH UP AN INCLINE (AASTU FINAL)

A 0.250 kg block is pressed 10.0 cm against a spring ($k = 525$ N/m) at the bottom of a 30° incline, then released ($g = 9.8$ m/s²). How far up the incline does it travel from the release point (a) if the incline is frictionless, (b) if $\mu_k = 0.20$?

Spring energy: $\frac{1}{2}(525)(0.100)^2 = 2.625$ J.

(a) $2.625 = mgd\sin 30^\circ = (0.250)(9.8)(0.5)d \Rightarrow d = 2.14$ m.

(b) $2.625 = mgd(\sin 30^\circ + \mu_k \cos 30^\circ) = (2.45)(0.5 + 0.173)d \Rightarrow d = 1.59$ m.

2.6.6 Power

D POWER

The **rate of doing work** (or of transferring energy):

$$P_{\text{avg}} = \frac{W}{\Delta t}, \qquad P = \frac{dW}{dt} = \vec{F}\cdot\vec{v} = Fv\cos\theta$$

SI unit: **watt** (1 W = 1 J/s). 1 horsepower (hp) = 746 W. The **kilowatt-hour** (kWh) $= 3.6 \times 10^6$ J is a unit of **energy**, not power.

WORKED EXAMPLE 6: LIFTING POWER

A motor lifts a 500 kg load 20 m at constant speed in 25 s ($g = 10$). Find the power.

$W = mgh = 500 \times 10 \times 20 = 1.0 \times 10^5$ J → $P = 10^5/25 = 4000$ W = 4 kW ≈ 5.4 hp.

WORK AND ENERGY TRAPS

- The **normal force** and the **centripetal force** do **zero work** (they are perpendicular to the motion).
- Friction does **negative** work on a sliding object.
- Kinetic energy can never be negative; potential energy can (it depends on the reference level).
- "Work done by a conservative force around a closed path is non-zero" is **false**: it is always zero.
- **kWh is energy**, and the watt is power.

Exam Practice

13 QUESTIONS · LESSON 2.6

1 Hawassa University · Mid Exam

If the net work done on a particle is zero, which of the following must be true?

A) The velocity is zero
B) The velocity is unchanged
C) The particle is at rest
D) The speed is unchanged

2 Haramaya University · Mid Exam

A constant force $\vec{F} = (3\hat{i} + 4\hat{j})$ N acts on a particle that moves from P(2, 6) m to Q(14, 1) m. The work done is:

A) 56 J
B) 16 J
C) 36 J
D) 20 J

3 Arba Minch University · Mid Exam

Which of the following is a necessary condition for mechanical work to be done?

A) A force must be applied
B) A displacement must occur
C) There must be a component of the force along the displacement
D) All of them

4 Bahir Dar University · Mid Exam

Which of the following statements about work is WRONG?

A) Work is a scalar quantity
B) The dimensions of work are ML^2T^{-2}
C) The SI unit of work is the newton-metre (joule)
D) The work done by the normal force on a sliding box is always positive

5 Ambo University · Mid Exam

Two balls have the same mass, and one moves twice as fast as the other. The ratio of their kinetic energies (slow : fast) is:

A) 1 : 2
B) 1 : 4
C) 2 : 1
D) 4 : 1

6 Dilla University · Mid Exam 2012

If the momentum of a body is doubled, its kinetic energy becomes:

A) twice as large
B) four times as large
C) half as large
D) unchanged

7 Wachemo University · Mid Exam 2013

The work done by a conservative force along a closed path is always non-zero.

True / False

8 Ambo University · Mid Exam

Find the work needed to stretch a spring of force constant 125 N/m by 2 cm from its natural length.

9 Arba Minch University · Mid Exam

A pendulum bob of mass M and length L is released from rest at a height $L/4$ above its lowest point. Find its speed when it is at height $L/8$.

10 AASTU · Final Exam

A 6.00 kg block of ice starts from rest and moves along x with $x(t) = \alpha t^2 + \beta t^3$, where $\alpha = 0.200$ m/s² and $\beta = 0.0200$ m/s³. Find (a) its velocity at $t = 4.00$ s, (b) the net force on it at that time, and (c) the work done on it in the first 4.00 s.

11 AASTU · Final Exam

A 0.250 kg block is held against a spring ($k = 525$ N/m) compressed 10.0 cm at the bottom of a 30.0° incline and released ($g = 9.8$ m/s²). How far up the incline does it move from the release point (a) if the incline is frictionless, (b) if $\mu_k = 0.20$?

12 University of Gondar · Mid Exam

A ball is dropped from a height of 1.00 m. Each bounce loses one quarter of its kinetic energy. After how many bounces does it no longer rise above 0.25 m?

13 Model question

A 60 kg student runs up a staircase 4.0 m high in 8.0 s ($g = 10$ m/s²). Her average power output is:

A) 300 W **B)** 240 W
C) 30 W **D)** 2400 W

Answers & Explanations

1. D

The speed is unchanged

$W_{\text{net}} = \Delta K = 0$, so $|v|$ stays the same. The **direction** can still change (e.g. uniform circular motion), so the velocity is not necessarily unchanged.

2. B

16 J

$W = F_x \Delta x + F_y \Delta y = 3(12) + 4(-5) = 16$ J.

3. D

All of them

$W = Fd\cos\theta$: zero force, zero displacement or θ = 90° each give zero work.

4. D

The work done by the normal force on a sliding box is always positive

The normal force is perpendicular to the displacement, so its work is **zero**.

5. **B**

1 : 4

$K \propto v^2$: $\frac{1}{2}mv^2 : \frac{1}{2}m(2v)^2 = 1 : 4$.

---

6. **B**

four times as large

$K = p^2/2m$, so $K \propto p^2$: doubling p multiplies K by 4.

---

7. **False**

For a conservative force (gravity, spring), the work around **any** closed path is **zero**; that is one definition of "conservative".

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8. **0.025 J**

$W = \frac{1}{2}kx^2 = \frac{1}{2}(125)(0.02)^2 = 0.025$ J.

---

9. **v = ½√(gL)**

$Mg\frac{L}{4} = \frac{1}{2}Mv^2 + Mg\frac{L}{8} \Rightarrow v^2 = \frac{gL}{4}$.

---

10. **(a) 2.56 m/s; (b) 5.28 N; (c) 19.7 J**

(a) $v = 2\alpha t + 3\beta t^2 = 1.60 + 0.96 = 2.56$ m/s.

(b) $a = 2\alpha + 6\beta t = 0.40 + 0.48 = 0.88$ m/s² → $F = ma = 5.28$ N.

(c) Work–energy theorem: $W = \frac{1}{2}(6.00)(2.56)^2 - 0 = 19.7$ J.

---

11. **(a) 2.14 m; (b) 1.59 m**

Spring energy $= \frac{1}{2}kx^2 = 2.625$ J.

(a) $2.625 = mgd\sin 30^\circ \Rightarrow d = 2.625/1.225 = 2.14$ m.

(b) $2.625 = mgd(\sin 30^\circ + 0.20\cos 30^\circ) = 1.649d \Rightarrow d = 1.59$ m.

---

12. **5 bounces**

After each bounce the height is ¾ of the previous one: $h_n = (0.75)^n$ m.

$(0.75)^4 = 0.316$ m (still above 0.25 m), $(0.75)^5 = 0.237$ m (below). So after the **5th** bounce.

---

13. **A**

300 W

$P = mgh/t = 60 \times 10 \times 4.0/8.0 = 300$ W.

CHAPTER 2 · LESSON 2.7

Linear Momentum, Collisions and Centre of Mass

AFTER THIS LESSON YOU CAN

- Define momentum and impulse and use the impulse–momentum theorem
- Apply conservation of momentum to collisions and explosions
- Distinguish elastic and inelastic collisions
- Locate the centre of mass of a system of particles

2.7.1 Momentum and impulse

D LINEAR MOMENTUM

$$\vec{p} = m\vec{v} \qquad (\text{unit: kg} \cdot \text{m/s} = \text{N} \cdot \text{s})$$

A **vector** in the direction of the velocity. Newton's second law in its general form: $\sum \vec{F} = \dfrac{d\vec{p}}{dt}$.

D IMPULSE AND THE IMPULSE–MOMENTUM THEOREM

Impulse is a force acting over a time: $\vec{J} = \vec{F}_{\text{avg}}\,\Delta t$ (for a variable force, the **area under the F–t graph**).

$$\vec{J} = \Delta\vec{p} = m\vec{v}_f - m\vec{v}_i$$

E WHY AIRBAGS AND PADDED FLOORS WORK

For the same change in momentum, a **longer collision time** means a **smaller average force**: $F = \Delta p/\Delta t$. Airbags, crumple zones, bending your knees when you land and catching an egg by moving your hands back all **increase** Δt.

✎ WORKED EXAMPLE 1: TENNIS BALL (BAHIR DAR FINAL)

A 0.0600 kg tennis ball arrives at 50.0 m/s and leaves the racket at 40.0 m/s in the opposite direction. Find the impulse and the work done on the ball.

Take the incoming direction as positive: $J = m(v_f - v_i) = 0.0600(-40.0 - 50.0) = -5.40$ N·s (5.40 N·s, directed back toward the server's side).

$W = \Delta K = \tfrac{1}{2}(0.0600)(40.0^2 - 50.0^2) = -27.0$ J.

! MOMENTUM TRAPS

- An object moving with **uniform velocity** has **zero change** in momentum (and zero impulse on it).
- Momentum is a **vector**: a ball bouncing back with the same speed has $\Delta p = 2mv$, not zero.
- Doubling the speed **doubles** the momentum but **quadruples** the kinetic energy.

2.7.2 Conservation of linear momentum

THE LAW

If the **net external force** on a system is zero, its **total momentum stays constant**:

$$\sum \vec{p}_i = \sum \vec{p}_f \qquad m_1\vec{v}_{1i} + m_2\vec{v}_{2i} = m_1\vec{v}_{1f} + m_2\vec{v}_{2f}$$

This follows from Newton's third law: during a collision the internal forces are equal and opposite, so the momentum lost by one body is gained by the other. It holds in **every collision and explosion** (the impact forces are so large that outside forces such as gravity and friction can be ignored during the short impact).

2.7.3 Types of collision

Type	Momentum	Kinetic energy	Example
Elastic	conserved	**conserved**	billiard balls (nearly), atoms and molecules
Inelastic	conserved	**not** conserved (some becomes heat, sound, deformation)	a car crash
Perfectly inelastic	conserved	**maximum** loss; the bodies **stick together**	clay balls, a bullet embedding in a block

COLLISION FORMULAS

- **Perfectly inelastic:** $v_f = \dfrac{m_1 v_{1i} + m_2 v_{2i}}{m_1 + m_2}$
- **1D elastic** (target at rest, $v_{2i} = 0$): $v_{1f} = \dfrac{m_1 - m_2}{m_1 + m_2} v_{1i}$, $v_{2f} = \dfrac{2m_1}{m_1 + m_2} v_{1i}$
- **Equal masses, elastic, head-on:** the balls **exchange velocities** (a moving ball stops dead and the other moves off with its speed).
- **Explosion from rest** (recoil): $0 = m_1 v_1 + m_2 v_2$, so the pieces fly apart in opposite directions with $v \propto 1/m$ (a gun recoils slowly because it is massive).

WORKED EXAMPLE 2: PERFECTLY INELASTIC COLLISION (ARSI EXAM)

A 6 kg body moving at +10 m/s collides head-on with a 4 kg body moving at −10 m/s, and they stick together. Find the final velocity and the kinetic energy lost.

$v_f = \dfrac{6(10) + 4(-10)}{10} = +2$ m/s.

$K_i = \tfrac{1}{2}(6)(100) + \tfrac{1}{2}(4)(100) = 500$ J; $K_f = \tfrac{1}{2}(10)(2)^2 = 20$ J → **480 J lost** (96 %).

WORKED EXAMPLE 3: COLLISION IN 3D (AASTU FINAL)

A 0.5 kg sphere with $\vec{v}_{1i} = (2\hat{i} - 3\hat{j} + \hat{k})$ m/s hits a 1.5 kg sphere with $\vec{v}_{2i} = (\hat{i} + 2\hat{j} - 3\hat{k})$ m/s. Afterwards the first moves at $(-\hat{i} + 3\hat{j} + 8\hat{k})$ m/s. Find the velocity of the second.

Total momentum: $0.5(2,-3,1)+1.5(1,2,-3)=(2.5,1.5,-4)$ kg·m/s.
$1.5\,\vec{v}_{2f}=(2.5,1.5,-4)-0.5(-1,3,8)=(3,0,-8)$ → $\vec{v}_{2f}=(2\hat{i}-5.33\hat{k})$ m/s.

2.7.4 Centre of mass

D CENTRE OF MASS (CM)

The point that moves **as if all the mass of the system were concentrated there** and all external forces acted on it. For a system of particles:

$$x_{cm}=\frac{\sum m_i x_i}{\sum m_i},\qquad y_{cm}=\frac{\sum m_i y_i}{\sum m_i},\qquad z_{cm}=\frac{\sum m_i z_i}{\sum m_i}$$

For a uniform symmetric object (rod, disc, sphere), the CM is at the **geometric centre**. The CM can lie **outside** the material (a ring, a boomerang).

- The CM moves according to $\sum \vec{F}_{\text{ext}}=M\vec{a}_{cm}$, and $\vec{p}_{\text{total}}=M\vec{v}_{cm}$.
- If there is no external force, the CM moves at **constant velocity** (or stays at rest), even if the system explodes. After a shell explodes in flight, its CM continues along the original parabola.

✎ WORKED EXAMPLE 4: FOUR MASSES (DILLA EXAM)

Masses 1 kg at (0, 0), 2 kg at (1, 0), 3 kg at (0, 2) and 4 kg at (1, −2) cm. Find the CM.

$M=10$ kg. $x_{cm}=\dfrac{0+2+0+4}{10}=0.6$ cm; $y_{cm}=\dfrac{0+0+6-8}{10}=-0.2$ cm. CM = (0.6, −0.2) cm.

! EXAM-BANK CORRECTION

One version of the AASTU five-mass centre-of-mass question gives (0.25, −0.25) m as the key. Recomputing with the given masses and positions gives **(0, −0.5) m**. Always recompute: $\sum m_i x_i/\sum m_i$ for each axis separately.

Exam Practice

13 QUESTIONS · LESSON 2.7

1 Wachemo University · Mid Exam 2013

If the total kinetic energy of a system is conserved, the collision is an inelastic collision.

True / False

2 Wolaita Sodo University · Mid Exam

In which type of collision are both momentum and kinetic energy conserved?

A) Elastic
B) Inelastic
C) Perfectly inelastic
D) All of the above

3 Arsi University · Final Exam 2014

Two pieces of clay collide and stick together. Which statement is true?

A) Both momentum and kinetic energy are conserved
B) Momentum is conserved but mechanical energy is not
C) Mechanical energy is conserved but momentum is not
D) Neither is conserved

4 Arsi University · Mid Exam 2014

A 6 kg object moving at $10\hat{i}$ m/s collides with a 4 kg object moving at $-10\hat{i}$ m/s, and they stick together. Their common velocity is:

A) $10\hat{i}$ m/s
B) $2\hat{i}$ m/s
C) $-2\hat{i}$ m/s
D) 0

5 Addis Ababa University · Final 2021

Two objects of unequal mass move toward each other with equal and opposite velocities and collide elastically. After the collision:

A) Both stop
B) The more massive object has the greater speed
C) The less massive object has the greater speed
D) They move together

6 Ambo University · Mid Exam

If the speed of an object doubles, its momentum and kinetic energy change by factors of:

A) 2 and 2
B) 4 and 2
C) 2 and 4
D) 4 and 4

7 Hawassa University · Mid Exam

What is the change in momentum of an object moving with uniform velocity between any two points?

8 Bahir Dar University · Mid Exam

An 80 kg skater starts from rest and reaches 10.0 m/s east. Find (a) her momentum at rest, (b) her final momentum, and (c) the impulse she received.

9 Bahir Dar University · Final Exam

A 0.0600 kg tennis ball moving at 50.0 m/s is hit back in the opposite direction at 40.0 m/s. Find (a) the impulse delivered by the racket, (b) the average force if the contact lasts 0.0100 s, and (c) the work done on the ball.

10 AASTU · Final Exam

A 0.5 kg sphere with velocity $(2\hat{i} - 3\hat{j} + \hat{k})$ m/s collides with a 1.5 kg sphere with velocity $(\hat{i} + 2\hat{j} - 3\hat{k})$ m/s. After the collision the 0.5 kg sphere moves at $(-\hat{i} + 3\hat{j} + 8\hat{k})$ m/s. Find the final velocity of the 1.5 kg sphere.

11 Dilla University · Mid Exam 2012

Four masses lie in the xy-plane: 1 kg at (0, 0), 2 kg at (1, 0), 3 kg at (0, 2) and 4 kg at (1, −2), all in cm. Find the centre of mass.

12 Ambo University · Mid Exam

Find the centre of mass of 1 kg at (1, 4, 4) m, 2 kg at (−2, 4, 1) m and 3 kg at (−1, 2, 2) m.

13 AASTU · Mid Exam

Three particles sit at the corners of a right triangle with an 8 m base and a 10 m hypotenuse: 4 kg at (4, 0) m, 2 kg at (12, 0) m and 2 kg at the top corner directly above the 2 kg base mass. Find the centre of mass.

Answers & Explanations

1. False

Conserving kinetic energy makes it **elastic**. In inelastic collisions kinetic energy is lost (momentum is conserved in both).

2. A

Elastic

Momentum is conserved in all three; kinetic energy only in elastic collisions.

3. B

Momentum is conserved but mechanical energy is not

Sticking together = perfectly inelastic: momentum is conserved, but kinetic energy is lost to heat and deformation.

4. B

$2\hat{i}$ **m/s**

$v_f = \dfrac{6(10) + 4(-10)}{10} = 2$ m/s in the $+x$ direction.

5. C

The less massive object has the greater speed

Both momentum and kinetic energy are conserved; the lighter object rebounds faster (the same total momentum is shared with $v \propto 1/m$ in the CM frame).

6. **C**

2 and 4

$p \propto v$ (× 2) and $K \propto v^2$ (× 4).

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7. **Zero**

Uniform velocity means $\vec{v}$ is constant, so $\vec{p} = m\vec{v}$ is constant and $\Delta\vec{p} = 0$ (no net force acts).

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8. **(a) 0; (b) 800 kg·m/s east; (c) 800 N·s east**

$p = mv$: (a) 0; (b) $80 \times 10.0 = 800$ kg·m/s east; (c) $J = \Delta p = 800$ N·s east.

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9. **(a) 5.40 N·s (opposite to the incoming direction); (b) 540 N; (c) −27.0 J**

(a) $J = 0.0600(-40.0 - 50.0) = -5.40$ N·s.

(b) $F = J/\Delta t = 5.40/0.0100 = 540$ N.

(c) $W = \Delta K = \frac{1}{2}(0.06)(1600 - 2500) = -27.0$ J.

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10. **(2î − 5.33k̂) m/s**

$$\vec{v}_{2f} = \frac{(2.5, 1.5, -4) - (-0.5, 1.5, 4)}{1.5} = \frac{(3, 0, -8)}{1.5} = (2, 0, -5.33) \text{ m/s}.$$

---

11. **(0.6, −0.2) cm**

$x_{cm} = (0 + 2 + 0 + 4)/10 = 0.6$ cm; $y_{cm} = (0 + 0 + 6 - 8)/10 = -0.2$ cm.

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12. **(−1, 3, 2) m**

$M = 6$ kg. $x = (1 - 4 - 3)/6 = -1$; $y = (4 + 8 + 6)/6 = 3$; $z = (4 + 2 + 6)/6 = 2$.

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13. **(8, 1.5) m**

Height $h = \sqrt{10^2 - 8^2} = 6$ m, so the top mass is at (12, 6).

$$x_{cm} = \frac{4(4) + 2(12) + 2(12)}{8} = 8 \text{ m}; \quad y_{cm} = \frac{0 + 0 + 2(6)}{8} = 1.5 \text{ m}.$$

CHAPTER 2 · QUICK REVISION

Kinematics and Dynamics of Particles — at a glance

CHAPTER 2 FORMULA SHEET

Topic	Formulas
Velocity, acceleration	$\vec{v} = d\vec{r}/dt,\ \vec{a} = d\vec{v}/dt$
Constant a	$v = v_0 + at,\ \Delta x = v_0 t + \frac{1}{2}at^2,\ v^2 = v_0^2 + 2a\Delta x,\ \Delta x = \frac{v_0+v}{2}t$
Free fall	$a = -g;\ H = v_0^2/2g;\ T = 2v_0/g;\ v = \sqrt{2gh}$
Projectile (level)	$T = \frac{2v_0 \sin\theta}{g},\ H = \frac{v_0^2 \sin^2\theta}{2g},\ R = \frac{v_0^2 \sin 2\theta}{g},\ \tan\theta = 4H/R$
Newton	$\sum \vec{F} = m\vec{a} = d\vec{p}/dt;\ W = mg;\ \vec{F}_{AB} = -\vec{F}_{BA}$
Friction	$f_s \le \mu_s N,\ f_k = \mu_k N$; incline: $a = g(\sin\theta - \mu_k \cos\theta)$
Circular motion	$a_c = v^2/r = \omega^2 r,\ v = 2\pi r/T,\ F_c = mv^2/r$
Gravitation	$F = Gm_1m_2/r^2,\ g = GM/R^2,\ v_{\text{orb}} = \sqrt{GM/r},\ T^2 \propto r^3$
Work, energy	$W = Fd\cos\theta,\ W_{\text{net}} = \Delta K,\ K = \frac{1}{2}mv^2,\ U = mgh,\ U_s = \frac{1}{2}kx^2$
Power	$P = W/t = Fv;$ 1 hp = 746 W
Momentum	$\vec{p} = m\vec{v},\ \vec{J} = \vec{F}\Delta t = \Delta\vec{p},\ \sum\vec{p}_i = \sum\vec{p}_f$
Centre of mass	$x_{cm} = \sum m_i x_i / \sum m_i$

TEN THINGS EXAMINERS LOVE

1. Distance ≥ |displacement|; average speed ≠ |average velocity|.
2. $a = 0$ means constant velocity, not rest; $v = 0$ at the top, but $a = g$.
3. Mass does not affect free fall (no air).
4. Projectile: v_x constant; $v_y = 0$ at the top; 45° gives the maximum range; complementary angles give equal ranges.
5. Action–reaction pairs act on **different** bodies.
6. Friction $= \mu N$, and N is not always mg.
7. UCM is accelerated motion, toward the centre.
8. Gravity: inverse square; orbit speed and period do not depend on the satellite's mass.
9. Normal and centripetal forces do no work; conservative work around a loop = 0.
10. Momentum is conserved in **all** collisions; kinetic energy only in **elastic** ones.